Muhammad Shair

Date of birth: 19/02/2003 | **Nationality:** Pakistani | **Phone number:** (+92) 3420295719 (Mobile) | **Email address:** shaair9211@gmail.com | **Website:** https://shaair9211.github.io/ | **LinkedIn:** Mohammad Shaair |

Whatsapp Messenger: +92-342-0295719 |

Address: R 632, Green Park City, National Highway, 75120, Karachi, Pakistan (Home)

WORK EXPERIENCE

GRADUATE TRAFFIC ENGINEER - (FULL TIME) – TRANSURBAN CONSULTING ENGINEERS – 02/09/2024 – 31/05/2026 – KARACHI, PAKISTAN

Areas of Technical Practice:

- Traffic and parking impact studies
- Traffic modelling
- Mechanical car stacker, car lift and car turntable design
- Construction certification
- Occupation certification
- Signage and line marking design
- Car parking and loading dock facilities design
- Construction traffic management plans
- Traffic control plans/Traffic guidance schemes
- Green travel plans
- Transport access guides
- Car park management plans
- Loading dock management plans

INTERN – MARHAMA GROUP LLC. – 09/03/2023 – 24/03/2023 – KARACHI, PAKISTAN

Practical Experience:

- Hands-on experience in quantity surveying and cost engineering, focusing on accurate estimation and budgeting.

Software & Estimation Skills:

- Utilized RFMS Measure and other estimation software to perform precise flooring estimations.

TRAFFIC ENGINEERING INTERN – DIGP TRAFFIC OFFICE – 14/09/2022 – 31/10/2022 – KARACHI, PAKISTAN

Practical Experience:

- Large-scale data analysis (including accident hotspot projection) using specialized tools.
- Used Python for data cleaning, handling, filtering & transformation.
- Data analysis to identify high-risk areas and understand accident patterns.
- Proposed actionable changes to road design and maintenance practices to mitigate risks and improve traffic flow.
- Developed sophisticated dashboards using Power BI and Excel to visualize data insights.

RESEARCH INTERN – GIS LAB, NED-UET – 09/03/2022 – 20/04/2022 – KARACHI, PAKISTAN

Geo-database Development:

- Contributed to developing a geo-database of road traffic injuries using GIS tools for data management and analysis.

Data Analysis:

- Worked in a team to analyze spatial data to uncover insights and support decision-making for traffic safety initiatives.

EDUCATION AND TRAINING

28/10/2020 – 12/07/2024 Karachi

BE CIVIL ENGINEERING NED University of Engineering and Technology

Website https://www.neduet.edu.pk/ | **Level in EQF** EQF level 7

30/08/2018 – 13/03/2020 Karachi, Pakistan

HIGH SCHOOL Govt. National College

SKILLS

Professional Competencies

Traffic analysis | Traffic modelling | Parking & access planning | Traffic control layouts | Construction traffic planning | Signage & line marking | Sustainable transport planning | Intelligent Transportation Systems (ITS) | Computer Vision for Traffic Applications | ITS Data Analysis | Traffic data collection & validation | Basic Electronics & Microcontrollers

Engineering Softwares

SIDRA Intersection | AutoCAD | AutoTURN | PTV Vissim / Visum | CoppeliaSim | Git | Primavera P6 | Synchro | MS Office | Arduino IDE | Fusion 360 | RFMS Measure

Programming

Python | Matlab | HTML | CSS | MicroPython | FPGA (Basic)

Libraries & Frameworks

Tensorflow | Open CV | YOLO | OpenPyxl | Numpy | Scikit-Learn | Pandas | Python-NEAT | Matplotlib | Playwright

PUBLICATIONS

2024

EXAMINATION OF THE EFFECT OF ILLEGAL PARKING ON CAPACITY REDUCTION OF URBAN ROADS (CSCE'24)

The study investigated the impact of illegal parking on traffic congestion, a major contributor to air pollution and emissions. Surveys and impact studies were conducted to assess the effect on the subject site. The study emphasizes the need for improved traffic management and recommends actionable changes to mitigate these issues.

2024

REAL-TIME SIGNAL OPTIMISATIN USING MACHINE LEARNING AND COMPUTER VISION

European Transport (under review)

The study evaluated the practicality of data-driven methods for real-time signal optimization. We developed an image processing model for real-time vehicle detection and counting and applied a machine learning model to optimize signal timing. The study was implemented at an intersection in Karachi, focusing on undisciplined traffic conditions.

COURSES

• Artificial Intelligence (Pakistan Freelance Training Program - PFTP)

• Python Introduction to Data Analysis and Machine Learning (Udemy)

• Bayesian Statistics & Supervised Learning (Udemy)

• Mindful Computing (Udemy)

• Oracle Primavera P6 (Udemy)

• Master Course in Smart Cities, Urban Planning & Development (Udemy)

PROJECTS

• Developed an image processing model to detect and count vehicles

• Designed a traffic trajectory plotter using computer vision

• Created a traffic intersection simulation using Python & Pygame

• Implemented a scheduling algorithm for real-time signal optimization using Python-NEAT

EXTRACURRICULAR

Co-Lead, IT Team (2022-2023)

IMechE - NEDUET Student Chapter

Member, "Lead All Department" Team (2021-2022)

IMechE - NEDUET Student Chapter

Participated in the workshop on "Towards Sustainable Solutions for Traffic-Related Problems in Karachi."

Winner of Hydrocast (Pervious Concrete Design Competition) at ACI-NEDUET

REFERENCES

Dr. Ashar Ahmed

Assistant Professor - NED University of Engineering and Technology
aahmed@neduet.edu.pk

Dr. Muhammad Ahmed

Assistant Professor - NED University of Engineering and Technology
muhammadahmed@neduet.edu.pk